

11. Carboxylic acids are classed as weak acids.

(a) State what is meant by the term weak acid.

[1]

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(b) Substituted propanoic acids,  $\text{CH}_3\text{CHXCOOH}$ , have different acid strengths. The table below gives some information about three substituted propanoic acids.

| Substituted propanoic acid,<br>$\text{CH}_3\text{CHXCOOH}$ |                                                                 |
|------------------------------------------------------------|-----------------------------------------------------------------|
| $\text{X} = \text{H}$                                      | The pH of a $0.50 \text{ mol dm}^{-3}$ aqueous solution is 2.59 |
| $\text{X} = \text{Cl}$                                     | $K_{\text{a}} = 1.48 \times 10^{-3} \text{ mol dm}^{-3}$        |
| $\text{X} = \text{OH}$                                     | $\text{p}K_{\text{a}} = 3.86$                                   |

Place these acids in order of increasing acid strength. **You must show your working.**

[3]

Weakest ..... Strongest



- (c) A buffer is produced by mixing  $100\text{ cm}^3$  of ethanoic acid ( $K_a = 1.8 \times 10^{-5}\text{ mol dm}^{-3}$ ) of concentration  $0.80\text{ mol dm}^{-3}$  with  $50\text{ cm}^3$  of aqueous sodium hydroxide of concentration  $0.80\text{ mol dm}^{-3}$  at a temperature of  $298\text{ K}$ .

- (i) Calculate the initial pH of the aqueous sodium hydroxide. [2]

pH = .....

- (ii) Calculate the pH of the buffer solution. [2]

pH = .....

- (iii) Explain how buffers such as this work. [3]

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